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# A Lightweight Noise-Robust Post-Trigger Classifier for Embedded Acoustic Artillery Launch Detection

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# Motivation – decide on the edge, in noise

- Passive acoustic sensing supports artillery early warning: it is **cheap**, needs **little power** and **emits no signal** – active systems are expensive, power-hungry and easier to locate.

## The decision has to be on the edge

- Cheap nodes mean many nodes, spread over a large area.
- Streaming raw audio from every node costs power and radio bandwidth – and a transmitting node is easier to find.
- So each node decides by itself and sends an **alert**, not the audio.

## The signal arrives weak

- Artillery fires from far away – in our recordings the guns were  $\approx$ **5–10 km** from the station.
- The launch reaches the sensor attenuated, on top of wind, traffic and other background.
- So the classifier has to work at **low SNR**.



**low SNR**

## Q1 Does it hold at low SNR?

Can a compact post-trigger classifier, trained with noise added to the **waveform**, stay reliable down to **5 dB** SNR?

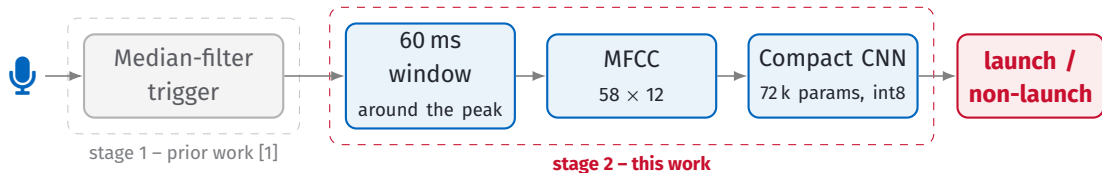


**on the edge**

## Q2 Does it survive the chip?

Does the resulting robustness survive **int8 quantization** and deployment on an **MCU** (ESP32-S3)?

# Two-stage detection pipeline



## Stage 1 – impulse detection

- Runs continuously on the stream: signal  $\rightarrow$  energy ( $x^2$ ), a **median filter** over neighbouring short windows estimates the local background.
- Background subtracted; a peak is a candidate only if it passes adaptive amplitude, RMS-ratio and energy tests and **risers like an impulse**.
- Detected all tested small-arms shots, reliable down to **5 dB SNR [1]**.

## Stage 2 – classification

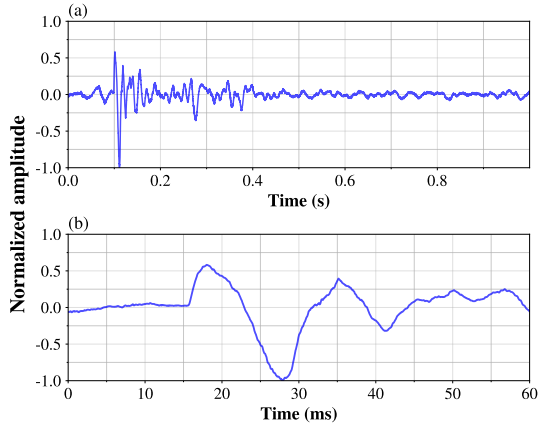
- Runs only on the candidates – peaks that look like an acoustic impulsive event.
- The trigger also fires on doors, shouts, small arms ... – a **binary** classifier rejects them.
- Launch / non-launch: stage 2 is in effect the **artillery launch detector**. We evaluate this stage only.

[1] J. Svatoš and J. Holub, "Smart Acoustic Sensor," in *Proc. 2019 IEEE 5th Int. Forum on Research and Technology for Society and Industry (RTSI)*, 2019, pp. 161–165, doi: 10.1109/RTSI.2019.8895591.

# Field recordings



- Two days, one **152 mm** self-propelled howitzer.
- Station 1-2 km from the impact area; guns  $\approx$ 5-10 km away.
- 4 synchronized mics (PreSonus PRM1), Roland Rubix 44; 16 bit, 22.05 kHz.



(a) launch recording; (b) the 60 ms analysis window cut around the detected peak.

# Labelled corpus and event-level split

| Subset                  | Events     | Samples    |            |            |
|-------------------------|------------|------------|------------|------------|
|                         |            | Train      | Val.       | Test       |
| <b>Artillery launch</b> | <b>50</b>  | <b>128</b> | <b>8</b>   | <b>10</b>  |
| Impulsive noise         | 706        | 457        | 109        | 140        |
| Small-arms shots        | 85         | 49         | 18         | 18         |
| Total                   | <b>841</b> | <b>634</b> | <b>135</b> | <b>168</b> |

- Non-launch: false-trigger candidates – door impacts, shouts, bells, **small-arms gunshots** as hard negatives.
- Split **by event**: the four channels of one shot never cross partitions.
- Training uses all 4 channels (launch share 20 %, class-weighted loss); validation and test use one.

# Small positive class – why, and how we handle it

## 💰 Why only 50 launches

- Artillery cannot be recorded on demand – only during a **live-fire exercise**, by arrangement with the military.
- The number of rounds was set by the exercise, **not by us**.
- Two recording days, one gun type.

## How we limit the impact

- **More views of each shot** – all 4 synchronized channels in training, plus one noisy copy per SNR level.
- **No leakage** – split by event, so no shot is in training and test.
- **Imbalance** – hard negatives and a class-weighted loss (1:4).
- **Against overfitting** – compact CNN (72 k), dropout 0.5, early stopping on a separate validation set.
- **Honest numbers** – test set used once;  $5 \times 5$  seeds; MCC instead of accuracy.

Still: 10 test launches and one gun type – generalization to other calibres is untested.

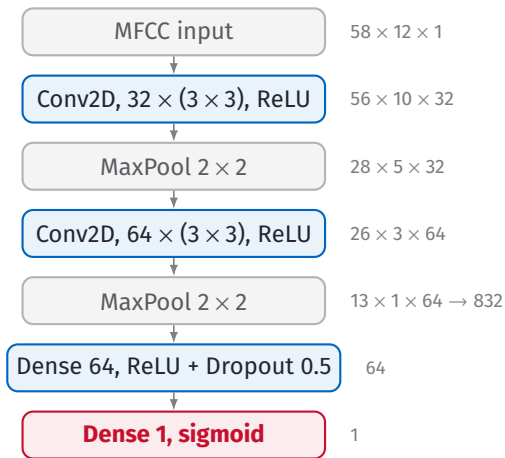
# MFCC front end and compact CNN

## MFCC

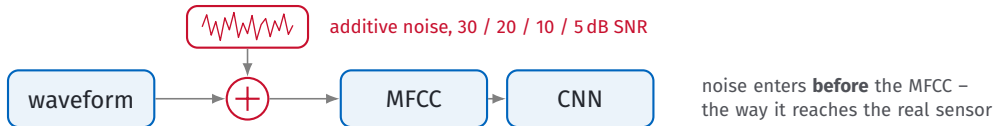
- 2.5 ms frames, 1 ms hop, Hamming
- 512-point FFT, 40 mel filters
- first 12 coefficients
- input tensor  $58 \times 12$

## Training

- Adam, binary cross-entropy
- early stopping on validation loss



72 193 trainable parameters



- **Expanded non-launch corpus** – small-arms shots and other impulsive sounds as hard negatives.
- **Class-weighted loss** for the 1:4 launch / non-launch imbalance.
- **Waveform-domain noise augmentation**, training split only – the MFCCs are recomputed for every noisy copy.

Control in the paper: the same noise added to the MFCC matrix instead does not give robustness (appendix).

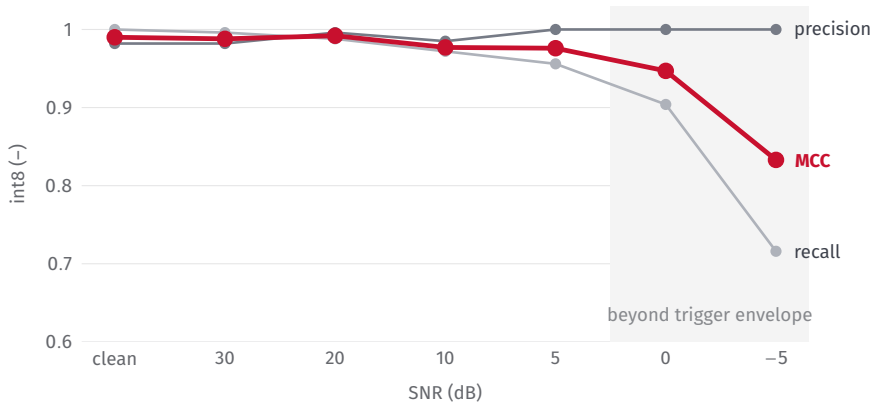
- Held-out test partition, used only for the final numbers.
- Noise added to the **waveform before MFCC extraction**, per event:

$$\sigma_n^2 = P_x \cdot 10^{-\text{SNR}/10}$$

at **30, 20, 10, 5 dB** – plus 0 and –5 dB as a stress test, beyond the trigger's 5 dB envelope.

- Peak detection and windowing re-run on the noisy signal.
- **5 training seeds** × **5 noise seeds** = 25 runs per cell, mean ± std; threshold 0.5.
- Three regimes: **float32** and **int8** on the PC, int8 on the **ESP32-S3**.
- Main metric: Matthews correlation coefficient (MCC).

## Result 1 – robustness at low SNR



Precision stays **1.000** from 5 dB down – in heavy noise the model misses launches, it does not raise false alarms. float32 vs.

int8: within  $\pm 0.02$  MCC, no consistent direction.

## Result 2 – on the ESP32-S3

### Same answers as the PC

**1171 of 1176** inferences bit-exact with PC int8.

The 5 others: scores at exactly 0.5 – a legal one-bit rounding difference between TFLite and TFLite Micro kernels.

### Latency

**32.0 ms** per MFCC segment (31.99–32.09 ms), independent of noise.

Classifier inference only – MFCC computed on the host.

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### TensorFlow Lite Micro, int8

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|              |          |
|--------------|----------|
| Model size   | 81 400 B |
| tensor_arena | 80 KiB   |
| Flash – code | 201.5 kB |
| Flash – data | 144.0 kB |
| RAM          | 143.8 kB |

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All 8 operators native in TFLite Micro; compute-heavy ones  
via ESP-NN kernels.

- **Stage 2 only** – no continuous-stream false-trigger rate; results assume the trigger found the peak.
- **Small positive class** – 50 launch events, one gun type (152 mm); other calibres untested.
- **Synthetic Gaussian noise** – added to the waveform, but no real field noise, propagation or device shift.
- **One event-level split**; no session-independent evaluation.
- **MFCC on the host** – on-device MFCC, end-to-end latency and energy per decision are future work.

- A compact MFCC-CNN (72 k parameters) works as the post-trigger stage of an embedded artillery launch detector.
- Trained with noise added to the waveform, it stays robust: **MCC 0.99 at 30 dB, 0.98 at 5 dB**; precision 1.0 under heavy noise.
- int8 is performance-neutral; ESP32-S3 matches the PC, 32 ms per inference.

Thank you – questions?

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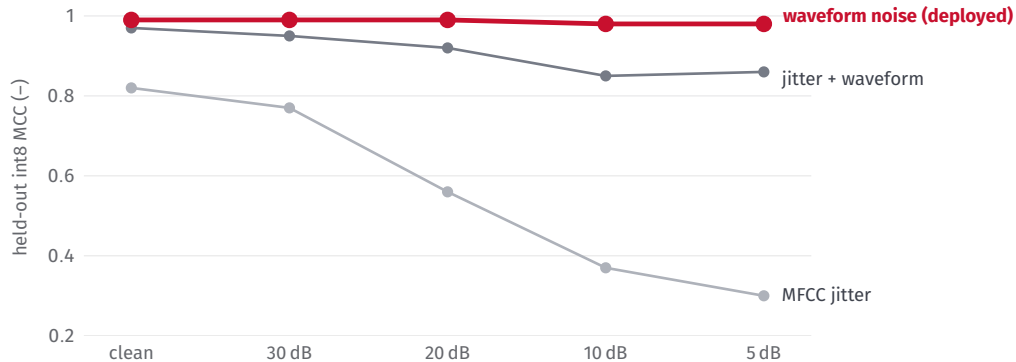
## Appendix – full held-out robustness table

| SNR (dB)        | Accuracy (%) | Precision     | Recall        | F1            | MCC                  | MCC (float32) |
|-----------------|--------------|---------------|---------------|---------------|----------------------|---------------|
| Clean           | 99.88 ± 0.24 | 0.982 ± 0.037 | 1.000 ± 0.000 | 0.990 ± 0.019 | 0.990 ± 0.020        | 1.000 ± 0.000 |
| 30              | 99.86 ± 0.26 | 0.982 ± 0.037 | 0.996 ± 0.020 | 0.988 ± 0.021 | <b>0.988 ± 0.022</b> | 0.998 ± 0.011 |
| 20              | 99.90 ± 0.22 | 0.996 ± 0.018 | 0.988 ± 0.033 | 0.992 ± 0.019 | 0.992 ± 0.020        | 0.993 ± 0.018 |
| 10              | 99.74 ± 0.39 | 0.985 ± 0.034 | 0.972 ± 0.061 | 0.977 ± 0.035 | 0.977 ± 0.035        | 0.974 ± 0.039 |
| 5               | 99.74 ± 0.42 | 1.000 ± 0.000 | 0.956 ± 0.071 | 0.976 ± 0.039 | <b>0.976 ± 0.039</b> | 0.978 ± 0.036 |
| 0 <sup>a</sup>  | 99.43 ± 0.53 | 1.000 ± 0.000 | 0.904 ± 0.089 | 0.947 ± 0.049 | 0.947 ± 0.049        | 0.934 ± 0.045 |
| -5 <sup>a</sup> | 98.31 ± 1.01 | 1.000 ± 0.000 | 0.716 ± 0.170 | 0.823 ± 0.122 | 0.833 ± 0.107        | 0.820 ± 0.088 |

Deployed int8 classifier, mean ± std over 5 training and 5 noise seeds. Test partition: 168 samples, 10 of them launches.

<sup>a</sup>Beyond the trigger's validated 5 dB envelope.

## Appendix – control: noise on the MFCC features



Same network, three augmentation strategies; mean over 25 runs. At 5 dB: jitter  $0.30 \pm 0.21$ , jitter + waveform  $0.86 \pm 0.07$ , waveform  $0.98 \pm 0.04$ .

Gaussian jitter on the precomputed MFCC matrix does not imitate noise that passes through the MFCC.

## Appendix – design choices

### Why supervised, not anomaly detection?

- After the trigger, *every* candidate is already anomalous against the background – small-arms shots included.
- One-class SVM / isolation forest / autoencoder would flag a rifle shot as “anomalous” – not the decision we need.
- They fit stage 1 (novelty detection) rather than stage 2.

### Why MFCC, not GTCC?

- GTCC won in our gunshot study (IMEKO 2026); MFCC is standardized with mature embedded implementations.

### Why 22.05 kHz?

- Launch energy lies well below the 11 kHz Nyquist limit.
- Buffers and MFCC compute scale with  $f_s$  – 44.1 kHz would spend RAM without adding usable signal.

### Why ESP32-S3?

- Low cost, low power, widely available; ESP-NN accelerated kernels for TFLite Micro.
- A representative Edge-AI MCU – not a claim about all embedded targets.

## Appendix – the five ESP32 disagreements

- All five: PC int8 score is **exactly 0.5** – on the decision threshold.
- The model is very confident, so the output quantization is coarse: **one LSB spans 10.8 logit units**.
- TFLite and TFLite Micro kernels may round one LSB differently – both legal.
- Platform behaviour is equivalent up to rounding at the boundary; 1171 / 1176 bit-exact.

**Precision** share of predicted launches that are real launches (false alarms).

**Recall** share of real launches that were detected (missed launches).

**MCC** Matthews correlation coefficient – uses the whole confusion matrix, robust to class imbalance; +1 perfect, 0 random.

$$\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

Test partition: 10 launches vs. 158 non-launch – accuracy alone would be 94 % for a model that never says “launch”.